

GREENWHEEL DBT FOUNDATION ASSIGNMENT

Quick-Start Analytics Engineering Workshop

1. The Problem Statement

The Scenario:

You have joined **GreenWheel**, a startup that rents electric bikes. Currently, our data is stored in simple CSV files. We need to move this data into our warehouse and create a clean "Analytics Layer" so our team can see which bikes are being used the most.

Your Goal: Learn the core dbt workflow: loading data, transforming it with SQL, and testing it for accuracy.

2. Step 1: Loading Data (Seeds)

Download the provided `raw_users.csv`, `raw_bikes.csv`, and `raw_trips.csv`. Place them in your `seeds/` folder and run:

```
dbt seed
```

3. Step 2: Creating Models

Build the following transformations using basic SQL. Focus on using the `{{ source() }}` and `{{ ref() }}` functions to link your models.

Model Name	Materialization	Requirement
stg_users	View	Select all users from the source. Convert names to Uppercase and emails to Lowercase.
stg_bikes	View	Select all bikes from the source. Rename model_type to bike_model for clarity.
fct_bike_usage	Table	Join trips with bikes (using ref). Calculate the total cost per bike and total minutes spent riding.

4. Step 3: YAML Configuration & Testing

Create a `schema.yml` file in your models directory. Add basic dbt "Generic Tests" to ensure our data is clean:

- **Unique & Not Null:** Apply these to `user_id`, `bike_id`, and `trip_id`.
- **Sources:** Define your `greenwheel` source so dbt knows where the seeds are.
- **Documentation:** Add a short description for each of your three models.

5. Final Checklist

Run these commands in order to verify your project:

```
dbt run
dbt test
dbt docs generate && dbt docs serve
```